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**Research Topic : A Systematic Literature Review on AI in
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A Systematic Literature Review on AI in Dentistry

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ABSTRACT

This systematic literature review explores the impact of artificial intelligence (AI) on dentistry, analyzing 58 papers from an initial pool of 550 sourced from IEEE Xplore, Scopus, Web of Science, and PubMed. The review categorizes the effects of AI in dentistry into several key areas: (i) integration and evolution of AI in dental diagnostics, (ii) personalized treatment planning, (iii) socioeconomic and demographic influences on AI models for treatment planning, (iv) patient autonomy and experience, (v) dental education (vi) the use of large language models (LLMs) in dentistry and (vii) impact of the dataset. The review highlights AI's role in enhancing diagnostic accuracy and efficiency, particularly through deep learning and neural networks, improving the detection of conditions such as caries and periodontal disease. It also examines AI's contributions to personalized treatment planning by analyzing extensive datasets, thereby reducing human error and improving patient outcomes. The review addresses the influence of socioeconomic and demographic factors on AI model effectiveness, emphasizing the need for diverse and representative datasets. It discusses AI's potential to enhance patient engagement and compliance through telemonitoring and oral hygiene tools improving communication between patients and dental professionals. In dental education, AI's incorporation prepares students and professionals for a technologically advanced practice environment with large language models like ChatGPT providing real-time advice and enhancing patient care and education. The review underscores the importance of balancing AI reliance with the development of critical analytical and decision-making skills in dental professionals. The analysis and discussion sections highlight challenges and ethical considerations in integrating AI into dental practice, including data privacy, algorithmic bias, and potential medical malpractice. The review emphasizes the need for robust policies and frameworks to ensure responsible AI development and calls for longitudinal studies to assess the real-world efficacy and long-term impact of AI-driven interventions on dental health outcomes.

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Keywords : Artificial Intelligence (AI), Dentistry, Dental Diagnostics, Personalized Treatment Planning, Neural Networks, Patient Engagement, Telemonitoring, Large Language Models.

I. Introduction : Background, Problem Analysis, Personal Motivation

Artificial Intelligence (AI) has extended its reach through various industries, offering transformative potential that has captivated researchers, practitioners, and policymakers alike. One such promising area is the integration of AI in dentistry, particularly in disease diagnosis and treatment planning. Dentistry has traditionally relied on manual experience-based diagnostics and treatment planning which is effective but can be time-consuming and subject to human error. Increasingly, dental professionals are incorporating AI tools into their practice, transforming diagnostic procedures, treatment planning, patient engagement, and education. Therefore, it is crucial to conduct this review to better understand the advancements, challenges, and impacts of AI in dentistry.

• Importance of Literature Review for Researchers, Practitioners, and Policymakers

This literature review serves as a valuable resource for researchers by consolidating and analyzing the latest advancements in AI applications within dentistry. By categorizing the effects of AI in areas such as diagnostics, personalized treatment planning, and dental education this review provides a clear roadmap of where the field currently stands and where it is headed. Researchers can use these insights to identify gaps in the existing literature and guiding future studies towards unexplored or under-researched areas. The review also highlights the impact of socioeconomic and demographic factors on AI model effectiveness and emphasizing the need for diverse and representative datasets.

For dental practitioners, this review offers practical insights into how AI technologies can be integrated into daily practice to enhance patient care. By examining the role of AI in improving diagnostic accuracy and efficiency, particularly through the use of deep learning and neural networks practitioners can better understand the potential benefits and limitations of these technologies. The review also discusses AI's contributions to personalized treatment planning which can help reduce human error and improve patient outcomes. AI's potential to enhance patient engagement and compliance through telemonitoring and oral hygiene tools is also explored providing practitioners with strategies to improve communication and patient satisfaction.

Policymakers will find this review beneficial as it outlines the ethical considerations and regulatory challenges associated with the deployment of AI in dentistry. The review addresses critical issues such as data privacy, algorithmic bias, and potential medical malpractice underscoring the need for robust policies and frameworks to ensure responsible AI development. By highlighting these challenges the review advocates for comprehensive guidelines that ensure AI systems are transparent, accountable, and used ethically. Policymakers can use this information to develop regulations that promote fairness in healthcare delivery and bridge the gap between different socioeconomic groups.

• Key Challenges and Debates in AI for Dentistry

The integration of AI in dentistry raises significant ethical concerns. Patient privacy, consent, and data security are paramount as AI systems often operate as "black boxes" with opaque decision-making processes. Ensuring that AI technologies are transparent and that patients understand how their data is used is crucial for maintaining trust. AI models are only as good as the data they are trained on. The review highlights the issue of variability and bias in AI models due to the limited availability of large, diverse datasets. This can affect the reliability

and fairness of AI recommendations, particularly for underrepresented populations. Addressing these biases is essential for developing equitable AI systems. There is a notable discrepancy in students' comprehension of patients' rights in dental practice for integrating AI. The review emphasizes the need for extensive education and training to help practitioners understand and effectively use AI tools. The potential for medical malpractice due to AI errors necessitates clear guidelines and protocols. The review underscores the importance of using AI as a support tool rather than a replacement for clinical judgment. Establishing legal and ethical frameworks to handle AI-related errors is crucial for protecting both patients and practitioners.

II. Methodology

- **This systematic literature review aims to answer the following key questions :**
 1. What are the latest advancements and methodologies in AI technologies for dental care, particularly in orthodontic diagnosis and treatment planning, and how have they contributed to improving patient outcomes?
 2. What are the main challenges and limitations associated with the use of AI in dentistry, including orthodontic care, and how do these impact the effectiveness and reliability of AI-based diagnostic and treatment planning tools?
 3. How do ethical considerations in the deployment of AI technologies in dentistry affect patient privacy, consent, and overall care, and what measures can be taken to ensure responsible use of AI in enhancing patient outcomes?
- **Data Sources and Search String:**

Four academic databases were used to search for scholarly publications for this review: IEEE Xplore, SCOPUS, ScienceDirect, Web of Science and PubMed. The Authors accessed the databases via the University of Ottawa Library, Canada. **IEEE Xplore:** A digital library providing access to technical literature in electrical engineering, computer science, and related fields, with extensive resources on AI technologies, **SCOPUS:** Contains comprehensive interdisciplinary research publications across various fields, **Web of Science:** A comprehensive research platform that indexes high-impact journals and conference proceedings across disciplines, useful for citation tracking and trend analysis in AI research, **PubMed:** The National Library of Medicine's search service providing access to MEDLINE, PREMEDLINE, and other related databases with links to participating online journals. The variety of subject content in these repositories is important because we wish to retrieve information from interdisciplinary sources to find a variety of viewpoints. Authors performed title and abstract searches in all four selected databases during the first month of the research, using the same search strings. The search expression was designed around concepts: advancement in artificial intelligence, AI in dentistry, clinical, and patient outcomes. The terms machine learning and deep learning were also included because they are heavily used in designing AI algorithms.

Covidence web-based software platform was used to streamline the production of this review including citation screening, abstract and full-text review, and extraction and export of data and references.⁹ This platform was used through the University of Ottawa’s institutional subscription.

((“Artificial Intelligence” OR “AI” OR “Machine Learning” OR “Deep Learning” OR “Neural Networks”) AND (“Dentistry” OR “Dental Care” OR “Orthodontics” OR “Orthodontic Diagnosis” OR “Teeth”) AND (“Patient Outcomes” OR “Clinical Outcomes” OR “Treatment Effectiveness” OR “Treatment Planning” OR “ETHIC” OR “ETHICAL”))

Figure 1 . Search Strings Designed for this Systematic Literature Review

- **Selection Criteria**

The selection of studies was guided by specific criteria to ensure the inclusion of relevant and high-quality research:

- **Relevance to AI in Dentistry:** Studies must specifically address the use of AI, machine learning, or related technologies within the field of dentistry. **Inclusion of Methodologies or Frameworks:** Prioritize studies that provide insights into specific AI methodologies or frameworks applied within dental care, offering a detailed examination of how AI technologies are implemented and their effectiveness. **Study Design:** Preference for peer-reviewed research articles, including reviews, meta-analyses, case studies, and clinical trials that provide empirical evidence on the effectiveness, challenges, or implementation strategies of AI in dental practices. **Publication Date:** To ensure the research is current and incorporates the latest advancements in AI, include studies published within the last 10 years. **Innovative AI Tools and Methodologies:** Studies that utilize or evaluate advanced AI tools and methodologies specific to dental care are preferred, reflecting the innovative approach highlighted in the research proposals. **Interdisciplinary Perspectives:** Favor studies that provide an interdisciplinary approach, integrating perspectives from computer science, dental science, ethics, and patient care, ensuring a comprehensive understanding of AI's impact on dentistry. **Addressing Bias and Fairness:** Include studies that explicitly tackle issues of bias, fairness, and accountability in AI algorithms and data sets used in dental diagnostics and treatment planning. **Language and Accessibility:** Studies must be available in English, and their content should be readily accessible for review, ensuring that the research can be thoroughly analyzed.

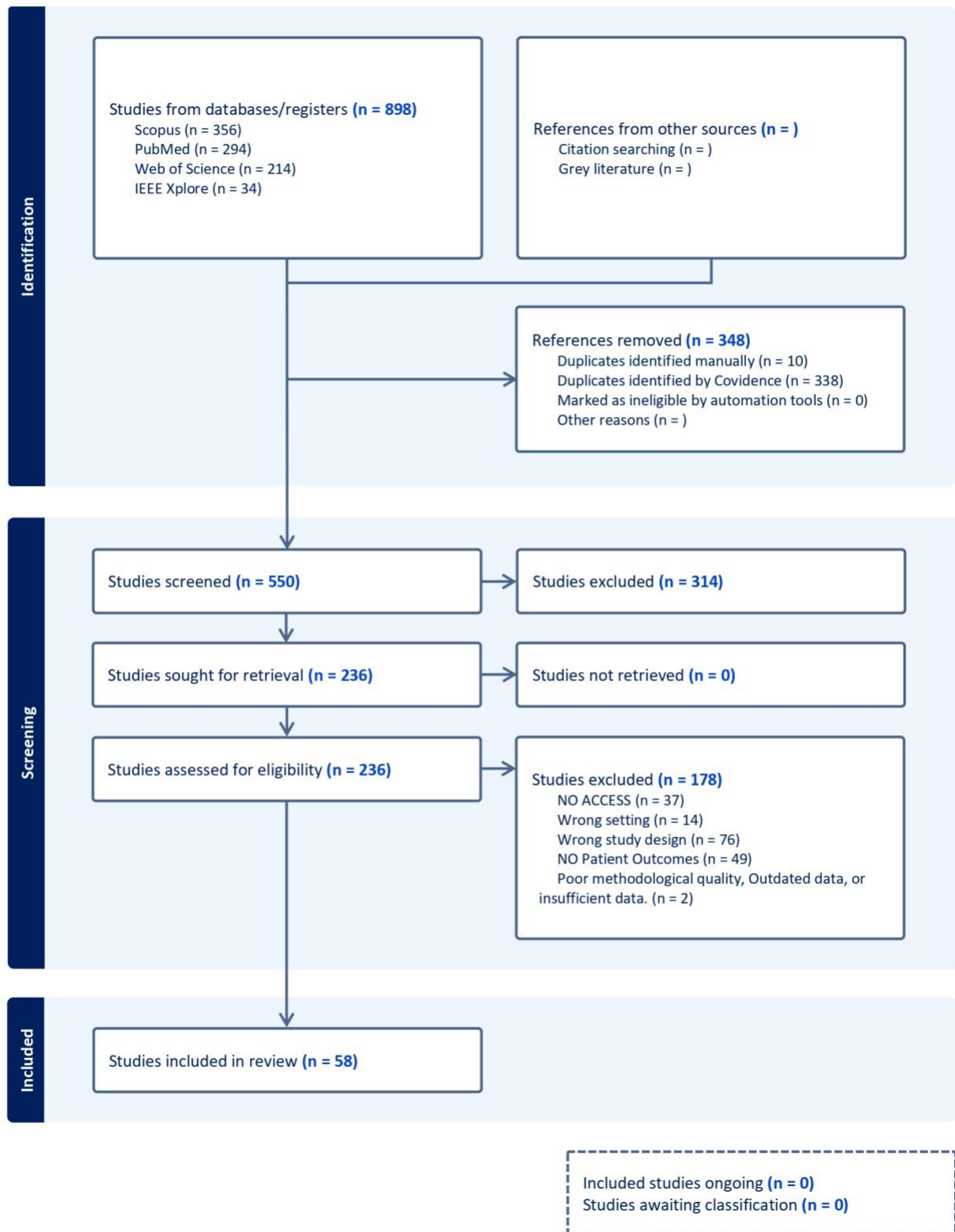


Figure 2: Studies Identified by the Search Strings and Selection Criteria.

III. Results

1. Integration and Evolution of AI in Dental Diagnostics

Dental diagnosis has significantly evolved with the integration of artificial intelligence (AI), enhancing the accuracy and efficiency of identifying dental diseases. Traditional diagnostic methods such as visual and tactical examinations combined with bitewing radiographs often suffer from low sensitivity and high rates of false positives and negatives when diagnosing caries [32]. AI models particularly deep learning (DL) and convolutional neural networks (CNNs) have shown superior performance in analyzing dental radiographs including bitewing, periapical, and panoramic images, to detect various dental conditions such as caries, periodontal bone loss, and periapical lesions [9] [33]. For instance, AI-based technologies have demonstrated higher sensitivity than dentists in detecting caries furcation involvement on dental radiographs with some models achieving accuracy rates as high as 94.59% and 92.5% in detecting cavities and 80% to 95 % for furcation Involvement [32] [39] [10].

a. AI Frameworks and Models in Dental Diagnostics.

AI frameworks like DeNTNet and DENTECT have been developed to analyze dental X-rays for periodontal bone loss and multiple dental procedures respectively showcasing the potential of AI in providing detailed diagnostic information and facilitating clinical evaluations [9]. DeNTNet, a deep learning-based technique focuses on detecting periodontal bone loss from dental X-rays and effectively identifies the affected tooth numbers based on the dental federation's notation although it is limited to periodontal bone loss detection only. On the other hand, DENTECT employs the YOLO model for object detection and is capable of recognizing five distinct dental procedures including periapical lesions, surgical extractions, conventional extractions, fillings, and root canal treatments presenting both speed and accuracy despite being trained on a relatively small dataset of 1005 images. Similarly, other studies have employed models like YOLOv6, YOLOv7, and YOLOv8, achieving improved detection capabilities and precision in identifying dental conditions such as cavities and tooth extractions [29][30][28][10][1][19].

b. Advanced AI Models for dental radiograph segmentation

The U-Net model particularly in its modified forms has been effectively employed for dental X-ray image segmentation achieving high accuracy and rapid processing times. The proposed CNN model for dental X-ray images which includes a U-Net architecture has been shown to process a large number of images quickly enabling faster decision-making by dentists and reducing patient waiting times. The CNN model achieved an accuracy rate of 98.60% indicating its effectiveness in classifying dental conditions from X-ray images [11]. The Teeth U-Net model is specifically designed for dental panoramic X-ray images. It addresses challenges such as variations in tooth shapes and overlapping structures. It achieved an accuracy of 96% [15]. Advanced approaches like the Dual-Hierarchy U-Net (DHU-Net) have also been developed to capture hierarchical structures of teeth, outperforming state-of-the-art methods in both whole tooth and internal tooth structure segmentation [26]. The segmentation and classification of individual teeth using a UNet architecture showed remarkable efficiency with AI models performing segmentation tasks over **1800 times** faster than human experts[37].

c. AI Systems for Detailed Image Analysis

AI systems, such as Diagnocat, leverage convolutional neural networks to perform detailed image analysis and segmentation providing comprehensive reports on dental conditions within minutes. It can significantly streamline the diagnostic process and improve treatment planning[6][40]. AI's ability to integrate diverse data domains including medical history and imagery allows for a more holistic understanding of patient health and potentially leading to more personalized and predictive dental care[38]. However, the field is still evolving, and the development of reliable AI models necessitates access to large, diverse, and well-annotated datasets, alongside the indispensable role of human expertise in interpreting AI-generated results[6].

The integration of AI in clinical practice has shown potential for cost-effectiveness as demonstrated by a study on AI for caries detection which reported higher sensitivity than dentists facilitating early detection and less invasive management of lesions. AI models have also been employed to recognize and assign tooth numbers in various radiographic images. A U-Net based convolutional neural network successfully generated clinical reports for caries detection, achieving a tooth numbering accuracy of 93% across tested radiographs enhancing the precision of dental charting and treatment planning [33].

Despite these advancements, the adoption of AI in daily dental practice faces challenges related to technology, sociology, and ethics, including limitations in data size, algorithm performance, and clinical application [32]. Sociologically, there is a notable lack of familiarity and trust among dental professionals regarding AI technologies. Many practitioners and students are not fully aware of AI's potential benefits and limitations, leading to hesitancy in its adoption. This is further exacerbated by concerns about AI replacing human elements in patient care, such as building rapport and counseling, which are crucial for successful treatment outcomes [20][34]. Ethically, the use of AI in dentistry raises significant concerns about patient autonomy, privacy, and fairness. AI systems often operate as black boxes making it difficult for patients and practitioners to understand the underlying decision-making processes. This lack of transparency can undermine trust and accountability necessitating robust ethical guidelines and regulatory oversight to ensure AI's responsible use in clinical settings [34][36]. Nonetheless, AI's ability to access large datasets and perform detailed analyses has made dental care more accessible and has provided healthcare practitioners with expert recommendations thereby improving diagnostic accuracy and patient outcomes [32]. The continuous development of AI in dentistry promises to address current limitations and further integrate AI into routine dental practice ultimately enhancing the quality of dental care.

2. Integration of AI in Personalised Treatment Planning.

AI algorithms can analyze extensive datasets, including patient records, dental images, and facial features, to create personalized treatment plans thereby improving diagnostic processes and reducing human error [2] [3]. In the realm of digital dental workflows, AI assists in the segmentation, localization, and measurement of anatomical structures on radiographic images which are crucial for planning oral and maxillofacial surgical procedures. For orthodontists who are relatively new to the practice or those seeking a second opinion, Artificial Neural Networks can be an invaluable resource. They offer a reliable auxiliary guide for treatment planning reducing the likelihood of clinical errors and improving treatment outcomes [4]. AI-powered systems such as those used in Digital Smile Design (DSD) enable dentists to visualize

and discuss potential treatment outcomes with patients fostering a more collaborative decision-making process and enhancing patient satisfaction [3].

a. AI in Predicting and Planning Orthodontic Extractions and Patterns.

The orthodontic treatment planning process is inherently complex requiring careful consideration of clinical data, personal knowledge, and the experience of the practitioner. This complexity is compounded by the need to make decisions on irreversible procedures such as tooth extractions which demand high concentration and expertise[14]. AI has shown promise in replicating near-clinician-level treatment planning by learning and processing both qualitative and quantitative data thus aiding in clinical decision-making [14]. Various AI models including Convolutional Neural Networks (CNNs), Random Forest Classifiers, and Artificial Neural Networks (ANNs), have been employed to improve the accuracy of cephalometric landmark identification and treatment planning [8] [13] [23] [26] [19] [47] [57]. The development of Clinical Decision Support Systems (CDSSs) using ML can further assist dental specialists in creating precise treatment plans, with studies showing an average accuracy of 84% in predicting treatment plans that align with those decided by orthodontists [18].

AI models have been used to predict the necessity of orthodontic extractions and their patterns incorporating multiple indexes such as arch length discrepancy (ALD) and molar key index, which are critical for comprehensive treatment planning [14]. ALD measures the difference between the space available in the dental arch and the space required to accommodate all teeth. This helps in identifying whether there is a need for space creation or extraction. A CNN based framework called DeAPIR significantly reduced computation time, requiring **only 64 seconds** to process a CBCT scan to calculate ALD [41]. Traditionally this decision relies on the clinician's expertise and diagnostic tests which can vary based on the practitioner's experience and clinical judgment [4]. Various AI models, including ResNet50, ResNet101, VGG16, and VGG19 have been employed for tooth extraction diagnosis using transfer learning and fine-tuning techniques. These models output a binary decision on the necessity of extraction with the highest AUC value of 0.961 achieved by the maxillary VGG19 model indicating high diagnostic accuracy [14].

A treatment planning system proposed in the paper [45], utilizes multi-layer perceptron neural networks and case-based reasoning to predict extraction needs and other treatment plans achieving success rates as high as 94% for extraction decisions. It also uses Case-based reasoning (CBR) which is a sophisticated approach in the field of artificial intelligence (AI) that focuses on solving new problems based on the solutions of previously solved, similar problems. This method has been adopted in various domains including orthodontics to enhance decision-making processes. CBR operates on the principle that similar problems have similar solutions. In the context of orthodontics, this means that the system can suggest treatment plans based on the outcomes of previous cases with comparable conditions.

AI-enhanced diagnostic tools have significantly increased the accuracy of treatment planning as evidenced by a randomized controlled trial where the AI group exhibited a mean discrepancy of 1.25 millimeters compared to 2.10 millimeters in the traditional group indicating a statistically significant improvement ($P < 0.001$) [7]. This precision in treatment planning has also resulted in **shorter treatment durations by 20%**, enhancing patient convenience.

Although the average cost is slightly higher than the traditional treatment cost. The development, maintenance, and operation of such sophisticated AI software require significant investment, which is reflected in the treatment costs but with wide adoption of AI tools it will drastically reduce [7]. AI's ability to analyze large datasets quickly and accurately allows for evidence-based decision-making, which leads to better patient outcomes and reduced medical errors.

b. AI in Pediatric Dental Diagnosis and Treatment Planning.

AI tools have also been applied to the segmentation of supernumerary teeth in pediatric patients. The paper [6] analyzed the potential benefits of artificial intelligence in the diagnostics of pediatric patients since cases of supernumerary teeth usually involve pediatric patients and having rapid diagnosis is crucial to reduce anxiety and efficient treatment plans. The commercial dental AI models Diagnocat and Relu were used which are based on convolutional neural networks (CNN) for automatic segmentation of the oral cavity and tooth structures [6]. Diagnocat was able to detect all the 9 supernumerary teeth from a scan of pediatric patient within minutes although Relu failed to detect the supernumerary teeth, it provided additional features such as the ability to refine the segmented structures, removing outliers, viewing the 3D scan in 2D cross sections, etc. Similarly, Diagnocat was also used in another study including a pediatric patient, It provided insights into the relationship between the root of the central supernumerary tooth (CST) and the additional impacted supernumerary tooth (IST), determining the easiest access point for extraction [21].

Another study [16] aimed to compare the accuracy and repeatability of cephalometric analysis results among three commercial AI-driven programs: CephX, WebCeph, and AudaxCeph. Significant discrepancies were found in specific measurements such as angle convexity and the occlusal plane angle, pointing to different methodologies used by the AI programs. The study emphasized the **need for standardization** across AI platforms to minimize discrepancies and the critical role of **clinician oversight** in evaluating automated results, especially for parameters with significant treatment implications. The discrepancy in results of all the models currently emphasizes the necessity to combine multiple tools to harness the potential of AI in dental radiology, indicating a gap in the development of a singular comprehensive AI tool for dental treatment planning.

c. Socioeconomic and Demographic Influences on AI Models for Treatment Planning.

The characteristics of a population used for training data, such as demographics, health behaviors, and genetic factors, can significantly influence the above mentioned model's predictions. If a model is trained on a specific population it may not perform well on a different population with different characteristics. One of the important characteristics is socioeconomic status which can impact health outcomes and access to healthcare. Models that do not account for these factors may **fail to generalize** to populations with different socioeconomic backgrounds [20]. Machine learning models like random forest and XGBoost have shown satisfactory performance in predicting tooth loss, with AUC values ranging from 0.827 to 0.906, highlighting the importance of incorporating socioeconomic factors alongside clinical data [17].

Also incorporating both clinical and demographic predictors such as age and smoking habits, a study used an ensembled model based on logistic regression and artificial neural network to predict the survival of molar teeth in patients with periodontal disease. To enhance the generalizability of the model the **dataset was collected from four different continents**, ensuring a diverse population sample. The ensembled model which combined logistic regression and neural networks demonstrated the best performance for predicting molar loss at the 10-year follow-up. This algorithm is now freely available for clinical use. It highlights the importance of considering socioeconomic and demographic factors in AI models for treatment planning as these influences significantly impact the accuracy and applicability of predictive models across varied patient populations[12].

It is important to note that AI systems cannot fully replace orthodontists in treatment plan preparation. Studies have shown that tools like Invisalign have an **accuracy of less than 50%** for all tooth movements and Clincheck Pro has been found to systematically underestimate tooth widths and show moderate reliability in estimating Bolton ratios which can **impact irreversible procedures** like interproximal enamel reduction [43].

The orthodontic profession has adopted digital technologies such as computer-assisted tooth movement and 3D simulations, and the next step involves creating neural networks based on general intelligence to suggest optimized treatment plans with more predictable outcomes [44]. The treatment planning process is complex and involves qualitative and subjective opinions making it challenging to build fully automated diagnosis algorithms.

3. Integration of AI in Patient Autonomy and Experience.

Patient autonomy and involvement in treatment planning are critical aspects of modern healthcare particularly in orthodontics and dentistry. AI and Big Data technologies offer promising solutions to enhance patient engagement and compliance addressing the 'last mile' problem in healthcare by improving accessibility and communication.

a. Enhancing Patient Engagement and Compliance

AI-based speech analysis using natural language processing (NLP) can support patient communication, making it easier for patients to understand their treatment plans and engage in decision-making processes [5]. This automation can help clinicians make more accurate and efficient treatment decisions thereby involving patients more effectively in their care plans. Two papers talked about accessing patient medical charts for treatment and prediction of oral health outcomes. The automated treatment planner proposed can reduce errors and examination time, allowing dentists to select more efficient treatment plans and report results with less error, which is particularly beneficial for novice dentists [30]. Patient charts whether paper-based or electronic are crucial for documenting oral health history and facilitating optimal care. AI can enhance these records by extracting meaningful data through ML algorithms, such as text mining and Latent Dirichlet Allocation (LDA), which can predict oral health outcomes based on patient chart data [42].

b. Streamlining Patient Scheduling

AI technologies can significantly enhance patient scheduling in dental clinics by streamlining the process and improving overall efficiency. AI-powered systems enable patients to schedule appointments online without the need for phone calls or manual coordination, integrating seamlessly with dental practice management software to check dentists' availability, suggest suitable time slots, and book appointments automatically, thus eliminating wait times and simplifying the scheduling process for both patients and dental practices[22]. One notable example is the use Dental Monitoring(DM) App [53] which can notify patients undergoing clear aligner treatment whether they need to schedule an appointment with the dentist or not by reading scans from portable dental monitoring scan boxes provided to patients. The use of such technology allows patients to provide necessary data to their dentist from the convenience of sitting at their homes.

c. AI in behavioural management of Pediatric Patients and Catering to Parents Needs.

Pediatric dentists often face challenges in managing children's behavior during dental visits which is crucial for effective treatment and fostering a positive attitude towards oral health. AI tools such as image analysis, natural language processing, and machine learning algorithms can be used for personalized oral hygiene education for pediatric patients[44]. These technologies facilitate better communication and education helping to alleviate fear and anxiety in young patients. Non-pharmacological techniques including effective communication, tell-show-do, and positive reinforcement, have been enhanced with AI-driven tools like virtual reality (VR) and augmented reality (AR) which serve as powerful distractions and engagement methods during dental procedures for children. Digital Smile Design allows for the customization of dental treatments based on the patient's unique facial and dental structure. For pediatric patients, this means treatments can be tailored to not only address current dental issues but also anticipate changes as the child grows. The use of DSD for creating simulated images of the smile could be particularly engaging for children and their parents making them more involved in the treatment process [2][3].

Furthermore, the ability to simulate, customize, and test various smile designs before proceeding with actual dental work ensures high accuracy and naturally aesthetic outcomes which are essential for patient satisfaction and case acceptance [39]. The integration of DSD into dental practice not only improves the aesthetic and functional aspects of dental restorations but also strengthens the relationship between the dentist, dental technician, and patient by facilitating better communication and understanding [3]. This collaborative environment is further enhanced by the use of cloud-based platforms like SmileCloud, which allow for the online storage and sharing of patient data, making the workflow more efficient and accessible from any location [3].

d. AI in Telehealth and Remote Monitoring

A Telehealth application requiring a dental monitoring scan box, retractors, and the patient's smartphone called Dental Monitoring (DM) has significantly streamlined the orthodontic treatment process by reducing the number of in-person appointments. Initially, patients with clear aligner treatment required three appointments in 10 weeks,

which DM has **reduced to two appointments** demonstrating a more efficient use of time and resources. This efficiency not only shortens the overall duration of treatment but also enhances patient satisfaction by minimizing disruptions to their daily lives. The use of the Digital Monitoring app has improved patient compliance by allowing for daily automated assessments of treatment progress. It facilitates direct communication between dental professionals and patients. Patients receive scan reports, messages, and reminders directly within the DM app making it easier to manage their treatment. This constant monitoring helps in identifying non-compliance early, leading to timely interventions and adjustments to the treatment plan, thereby ensuring better outcomes. Despite these improvements, DM does not necessarily decrease the average number of appointments per year for Dentists. This is because the AI technology used in DM can quickly identify issues such as dropped attachments or damage to orthodontic accessories from patients' home video scans necessitating additional visits to the dental office for repairs[53].

e. AI in Oral Hygiene Education.

The integration of AI in oral hygiene education has demonstrated significant potential, particularly through the use of intelligent toothbrushes and sensors that offer real-time feedback on toothbrushing techniques and habits. These technologies aim to improve oral hygiene by identifying areas of dental biofilm accumulation and gingival inflammation thereby providing actionable feedback for users to enhance their brushing practices. However several challenges persist in the long-term compliance of users and the quality of intra-oral images captured by these devices. Studies have shown that while user feedback on these AI-assisted devices is generally favourable there is a lack of qualitative data on long-term usage and compliance with only one study extending beyond a year. The quality of intra-oral images particularly those of the posterior, palatal, and lingual surfaces remains a significant issue limiting the effectiveness of these tools as comprehensive screening devices for home use. The high cost of these AI devices also poses a barrier to widespread adoption, especially among the elderly and socio-economically disadvantaged populations who could benefit the most from increased access to dental care [46].

4. Integration of AI in Dental Education.

The successful integration of AI requires a favorable environment for dental professionals to become familiar with utilizing AI across all dental disciplines along with ongoing education and training. Current curricula in dental education are increasingly incorporating AI training to prepare students and professionals for the evolving landscape of dental practice. For instance, the integration of Virtual Reality (VR) simulation technology into preclinical dental education is a notable example. VR simulators like Simodont and the Virteasy Dental Haptic Simulator are used to provide students with hands-on experience in clinical scenarios enhancing their tactile skills through haptic technology which simulates the sense of touch by applying force vibrations or motion to the user [40].

a. Perceptions and Attitudes Towards AI

Ethical considerations such as patient consent, data privacy, and algorithmic bias are paramount to ensuring that AI technologies are implemented responsibly and in a patient-centered manner [24]. Studies have shown that dental students generally have

positive perceptions regarding the ethical considerations of using AI in clinical decision-making. A study conducted in a Malaysian dental institute revealed that dental students had positive perceptions about the ethical use of AI with female respondents showing stronger agreement on the importance of patient consent and privacy prioritization, and final-year students scoring slightly higher overall [24]. In India, a cross-sectional study among undergraduate and postgraduate dental students found that 59.6% agreed AI could be used as a definitive diagnostic tool, 66.5% supported its use for radiographic diagnosis of tooth caries, and 71.3% believed AI could be a valuable treatment planning tool. Additionally, 55.7% of respondents felt AI should be part of undergraduate dental training [34]. These positive perceptions are crucial as dental students are the future practitioners who will integrate AI technologies into their professional careers and understanding their perspectives helps shape responsible and sustainable AI integration in dental practice.

Dental students generally have positive perceptions of AI's ethical considerations. But there is a lack of profound understanding of patients' rights particularly regarding access to and review of the algorithms used in their clinical decision-making. There was a notable deficiency in the **students' comprehension of patients' rights** in dental practice. This is particularly evident in their views on the necessity for patients to access and review the algorithms used in their treatment. The students did not see the importance of allowing patients to understand the data and logic behind the AI's clinical decisions [24]. Similar attitudes were observed in a study among medical students in Iran, where a significant portion exhibited low awareness of patient rights. This gap highlights the need for dental educators to integrate lessons about patient rights and ethics into the curriculum to prepare future practitioners for responsible AI integration[24].

Resistance within the dental education community towards integrating AI in dental education can stem from several factors. A significant barrier is the lack of awareness and understanding of AI's potential and limitations among both students and educators. Many dental students and professionals may misunderstand AI and view it as a threat to their jobs rather than a tool to enhance their capabilities [32][38]. This misconception can lead to apathy and resistance to adopting AI technologies [32]. One of the main reasons for resistance can be fear among dental students and professionals that AI might replace them, making their roles obsolete. This misconception can arise from not fully understanding that AI is intended to enhance, not replace, human capabilities in dentistry.

b. Proposed Curriculum for AI in Dental Education

A proposed curriculum for integrating AI in dental education [49] features an incremental stepwise approach across the program. It leverages a multidisciplinary team including clinicians, scientists, and bioethics professionals. Early in the curriculum students can be introduced to bioinformatics by visiting computer science lecturers and exploring medical datasets and relevant research articles. As they progress, students can analyze AI's performance in diagnostics and treatment planning and practice **communicating its applications to patients**. During the clinical phase, the curriculum emphasizes ethical and legal considerations of AI use engaging students in discussions about biases, misapplications, and the potential legal ramifications.

Through case studies and group discussions, students can critically examine issues like data privacy, algorithmic bias, and the reliability of AI tools.

5. Integrating Large Language Models (LLMs) in Dentistry: Opportunities, Challenges, and Ethical Considerations.

The integration of Large Language Models (LLMs) in dental education presents both opportunities and challenges that need careful consideration. LLMs, such as OpenAI's ChatGPT and Microsoft's Bing Chat have demonstrated potential in enhancing educational outcomes by providing accurate responses to dental exam questions and clinical guidelines with Bing Chat achieving a 71.99% accuracy rate in dental exams surpassing the passing threshold of 60% [48]. On the other hand, ChatGPT achieved a mean accuracy of 99.0%, significantly higher than senior students (79.7%) and junior students (77.0%) [50].

A study involving 354 respondents, including dental students and professionals revealed that more than 90% were willing to use ChatGPT for different dental tasks with a majority recognizing its potential in advanced dentistry, especially in dental education and research [52]. The integration of tailored ChatGPT in oral health can provide real-time advice on oral hygiene practices, answer queries about oral health issues, facilitate booking dental appointments, and encourage adherence to dental checkups, thereby offering a personalized and interactive approach to patient care [51][54].

These findings suggest that AI can significantly outperform dental students in terms of diagnostic accuracy, consistency, and data spread in endodontic assessments. The integration of AI into dental curricula must be approached with caution. There is a significant concern that dental students and recent graduates might become overly reliant on AI tools like ChatGPT. The "garbage in, garbage out" (GIGO) principle emphasizes that the accuracy and reliability of the information provided by ChatGPT depend on the quality of the input data it receives. If dentists or dental students input incorrect or misleading information, the responses from ChatGPT will reflect those inaccuracies[54]. This overreliance could potentially impede the development of their critical analytical and decision-making skills leading to a false sense of confidence. The heavy use of AI in diagnostic processes might overshadow the importance of developing and honing clinical judgment and critical thinking skills which are crucial for dental professionals [50].

6. Impact of Datasets and replication of human judgement of AI in Dental Practice.

The integration of synthetic datasets in AI for dental practice addresses several critical challenges associated with traditional data acquisition. Traditional datasets in dentistry often suffer from issues such as privacy concerns, extensive manual annotation requirements, and inherent biases, which limit their scalability and generalizability. Synthetic datasets generated using advanced generative AI techniques like variational autoencoders, generative adversarial networks, and diffusion models can overcome these limitations by producing diverse and representative data without compromising patient privacy. Synthetic datasets can mitigate ethical concerns related to data privacy and security, as they do not involve real patient data thereby ensuring compliance with legal and regulatory requirements. However, the reliability of synthetic data must be rigorously validated as issues like data hallucination and deepfakes can introduce errors that could be detrimental in clinical settings [55]. The Synthetic Dental Radiograph (SDR) dataset was used by researchers for a dental age estimation model with

specific variations in dental morphology and age labels, facilitating Controlled Diversity, Data Privacy, Bias Reduction, and Cost Effectiveness [56].

IV. Analysis and Discussion

The integration of artificial intelligence (AI) in dental diagnostics has significantly improved accuracy and efficiency, outperforming traditional methods. For instance, AI models like deep learning (DL) and convolutional neural networks (CNNs) excel in detecting disease by quickly analyzing radiographs and suggesting treatment plans. In resource-constrained settings like South Africa where dental screening centers are overwhelmed AI models can automate the analysis of panoramic radiographs, ensuring that only workable images are processed further thus optimizing the workflow and reducing the burden on radiologists.

AI fosters collaboration by visualizing potential outcomes and predicts orthodontic needs based on indexes like arch length discrepancy. AI tools also segment supernumerary teeth in pediatric dentistry although standardization across platforms is needed for reliability. Demographics and socioeconomic factors influence model effectiveness emphasizing the need for diverse training data. AI improves patient engagement, compliance, and oral hygiene education through telehealth applications like the Dental Monitoring App[53] and intelligent devices like smart toothbrushes[46]. AI training in dental education emphasizes ethics but misconceptions about AI replacing human roles persist. Large Language Models (LLMs) like ChatGPT enhance education and patient care by providing real-time advice although reliance on AI requires developing critical skills[51][54].

Using AI technologies in dentistry safely and effectively is paramount to achieving improved patient outcomes and maintaining trust in these innovative tools. Choosing AI models based on their ability to deliver accurate, efficient, and patient-friendly outcomes is crucial for improving global oral healthcare. By carefully selecting models that excel in these areas dental professionals can ensure better diagnostic accuracy and patient satisfaction. Addressing these challenges will fully harness AI's potential, improving dental diagnostics and patient care.

1. Personifying artificial intelligence (AI) in dentistry as a "Digital Assistant".

An interview with a Dentist/CEO of a dental healthtech AI startup provided valuable insights into the practical applications and considerations of AI as a “Digital Assistant” but not a replacement for human experts in dentistry. The dentist highlighted that ***“While AI significantly enhances the clarity and precision of 3D x-ray images, human expertise remains crucial for accurate diagnosis and treatment planning. For example, AI can adjust images or objects within them, making them more defined and easier to interpret, but the final interpretation and decision-making still rely heavily on the dentist’s clinical judgment.”***

It is an apt description, artificial intelligence models in dentistry can be described as a “Digital Assistant” that enhances the capabilities of dental practitioners by performing tasks traditionally associated with human intelligence. These AI systems can execute a variety of tasks with greater accuracy and minimal manpower ranging from appointment scheduling and coordination to aiding in clinical evaluation and therapy which was **traditionally performed by dental technicians or nurses**. This does not lead to a threat to their jobs but instead is just **augmented intelligence to enhance human capability** and intelligence. The inclusion of AI as a “digital assistant” in dentistry will transform the roles of dental technicians and nurses

emphasizing their importance in **bridging below gaps** between AI technology and clinical practice[53].

- a. **Accuracy Check:** Dental assistants and nurses will review AI-generated reports for accuracy ensuring that the data matches patient records and treatment progress. This step is crucial because AI can sometimes misinterpret data or overlook nuances that a trained human eye can catch.
- b. **Identifying Relevant Information:** They will sift through the information provided by AI to identify the most relevant and critical data for orthodontic treatment.
- c. **Reporting Errors or Anomalies:** When AI systems report errors, such as incorrectly dropped attachments or fail to track changes in treatment accurately. They ensure that such errors are corrected or flagged for further review by the orthodontist.
- d. **Enhancing Treatment Plans:** By filtering AI reports dental assistants and nurses will contribute to the refinement of treatment plans. They help in adjusting the plans based on AI insights combined with their professional judgment and experience.
- e. **Handling Complex Tasks:** Dental assistants and nurses can focus more on complex tasks that require **empathy, practical thinking, and physical engagement** moving away from routine and stereotypical tasks.
- f. **Patient Communication:** Dental assistants and nurses will play a key role in communicating the findings from AI reports to patients translating technical details into understandable language and advising them on the next steps in their treatment.

With the above mentioned roles, the most important role for dental assistants and nurses will be **ethical oversight**. They need to ensure that the use of AI adheres to ethical guidelines including patient consent and data privacy. This involves understanding the ethical implications of AI and advocating for patient rights.

2. Challenges in Replicating Human Judgment in Orthodontic AI Systems, Need for Individual Customization.

Moreover, replicating human judgment in automated systems, particularly in the domain of orthodontics presents a complex challenge due to the multifaceted nature of clinical decision-making. Human judgment in clinical settings is characterized by deep integration of empirical knowledge, experiential learning, and intuitive reasoning, making it a sophisticated process that is difficult to encapsulate within the rigid frameworks of artificial intelligence (AI) algorithms.

From the perspective of dentist/CEO at a dental healthtech AI startup, “**Ensure that AI tools used in treatment planning are customizable and adaptable to individual dentist preferences. This involves designing user interfaces that are intuitive and allow easy modifications of AI suggestions**”. The inherent subjectivity and variability in clinical decision-making, influenced by individual clinician's knowledge, experience, and even cultural perspectives add layers of complexity that automated systems struggle to mimic. One of the primary difficulties in

replicating human judgment lies in the nuanced understanding and interpretation of clinical data. Clinicians often make decisions based on subtle cues and patterns within the patient's data, integrating both quantitative measurements and qualitative assessments in their reasoning process. This level of nuanced interpretation is challenging for AI to replicate as it requires not only the detection of patterns within large datasets but also an understanding of the context and significance of these patterns.

3. Influence of medical and technological malpractice on integration of AI into Dentistry.

The potential for medical malpractice significantly influences the adoption of AI in dental practice. AI's role as a decision support tool rather than a replacement for clinical judgment is crucial, as errors in AI recommendations can lead to malpractice claims against dentists who fail to critically evaluate these suggestions.

From the perspective of dentist/CEO at a dental healthtech AI startup, ***“Medical malpractice in dentistry commonly involves issues such as delayed treatments, preferring aesthetic outcomes over complete dental treatment, surgical mistakes, use of improper tools, and complications arising from incorrect treatments.”*** It was highlighted, that one of the potential malpractice issues with the integration of AI can be an underestimation of the severity of dental disease and dentists completely relying on AI recommendations. Commercially available AI dental tools such as Diagnocat[6], Clincheck Pro[43], CephX[16] already provide warnings that the responsibility for actual treatment assessment lies with the dentist. The lack of established case law on AI use in dentistry further complicates accountability, with dentists potentially liable for errors resulting from improper AI system implementation or misuse[58].

Another recommendation from the dentist was ***“To address potential malpractice risks associated with AI, it is essential for dental professionals to understand when and how to rely on AI recommendations. Implementing guidelines and protocols for AI use in clinical settings that can help manage these risks.”*** Medical malpractice by dentists, particularly for patients seeking aesthetic improvements rather than comprehensive dental treatment, can significantly create hurdles for the effective integration of AI models in dental practice. Dentists may overlook or disregard the AI model's recommendations for complete and correct treatment plans to cater to patients' desires for more aesthetically pleasing outcomes. This not only compromises the quality of care but also undermines the credibility and reliability of AI systems. Additionally, if a dental AI model is designed to prioritize treatment plans that enhance aesthetic appeal over providing thorough and necessary dental care, it constitutes an ethical issue of **deception** in AI.

The implementation of AI in Digital Smile Design must be approached with caution to avoid dark patterns in AI. **Dark patterns**, which are design tactics that trick users into making decisions that benefit the service provider at the user's expense, can manifest in various ways in dental AI applications. For example, if an AI model is programmed to emphasize overly optimistic or unrealistic aesthetic outcomes for a child's smile it can manipulate parents into agreeing to unnecessary or more expensive treatments. By presenting an idealized simulation that prioritizes cosmetic appeal over the child's long-term dental health needs parents might be unduly influenced to pursue treatments that are not medically necessary. For dentists, they may feel compelled to adhere to the AI-recommended plans to maintain consistency with the visual presentations shown to the family.

Tailored large language model (LLM) tools used for patient engagement such as virtual assistants or chatbots, might use urgency or fear-based tactics to convince patients to consent to immediate extensive treatment plans. They might, for example, use phrases like "*Immediate action required to prevent further damage*".

AI applications in dentistry often require extensive data sets to train and refine their algorithms. In some instances, dental software might collect patient data subtly during routine procedures without explicit consent or clear disclosure about how this data will be used. This could include details about patients' dental conditions, treatment responses, and even personal identifiers that are then used to train AI systems or for purposes unrelated to the patient's immediate care such as targeted advertising or selling data to third parties.

To maintain ethical standards and ensure the responsible use of AI in dentistry it is crucial to design AI systems that balance aesthetic and functional dental needs, providing complete and accurate treatment plans without deceptive influences. This approach will foster trust in AI applications and uphold the integrity of dental practice.

4. Need for Longitudinal Studies on AI Effectiveness in dental practice.

The true impact of dental AI models on real-world clinical practice remains unclear as most studies have focused on internal testing rather than external validation which is crucial for assessing the generalizability of AI models across different clinical settings. The result section revealed that only one of the reviewed studies performed longitudinal research and discussions about patient outcomes were rarely addressed [7]. Additionally, AI may have demonstrated high sensitivity in detecting early caries and other dental pathologies, but its effectiveness in improving patient outcomes and reducing healthcare costs over the long term has not been thoroughly investigated.

Most studies focus on the immediate accuracy and efficiency of AI models often highlighting their superior diagnostic capabilities and potential to enhance clinical decision-making. However, understanding the true impact of AI on patient outcomes requires comprehensive longitudinal studies. Such studies would help in tracking patient health over extended periods, providing insights into the real-world efficacy of AI-driven interventions in improving dental health, reducing treatment times, and enhancing overall patient satisfaction. They would also help identify any long-term adverse effects or limitations of AI applications in dentistry that short-term studies might overlook.

5. Need for Ethical AI Development : Policies and frameworks.

The development of ethical AI systems necessitates robust policies and frameworks that address a plethora of ethical considerations. They are essential to guide the design and implementation of AI systems ensuring they align with human values and societal expectations. In our review, we found 4 papers highlighting the issue of ethical consideration in dental AI development. This includes the necessity of the Montreal Declaration and the European Union Ethics Guidelines for trustworthy AI.

The Montreal Declaration, in particular, underscores the importance of integrating ethical considerations such as **Well-being** : AI should promote the well-being of all sentient beings, Develop AI tools that prioritize patient health and comfort, **Respect for Autonomy** : Ensure patients are fully informed about the use of AI in their treatment and obtain their explicit

consent, **Protection of Privacy and Intimacy** : AI must guarantee the protection of privacy and respect the intimacy of individuals, **Solidarity**: Train AI models on diverse datasets to ensure fair and equitable treatment recommendations, **Democratic Participation**: Engage patients, clinicians, and other stakeholders in the development and deployment of AI systems, **Equity**: Regularly evaluate and adjust AI systems to ensure they do not perpetuate biases, **Diversity and Inclusion**: Develop AI tools that are sensitive to cultural differences and individual patient needs, **Prudence**: Maintain human oversight in AI-driven decision-making processes, **Responsibility**: Establish clear accountability structures for the development and use of AI in orthodontics [35].

The European Union's guidelines further elaborate on these principles, advocating for stringent data privacy and security measures such as **Data Protection and Cybersecurity**: Emphasizing the need to protect personal data and ensure robust cybersecurity measures to safeguard information, **Transparency**: The necessity for AI systems to be transparent, allowing users to understand how AI decisions are made. This includes explaining the accuracy of AI and its decision-making processes, **Accountability**: Highlighting the importance of accountability in AI systems, ensuring that there are measures in place to hold developers and users responsible for the AI's performance and outcomes.[31]

Often, there are overly optimistic assumptions about the readiness and infrastructure of healthcare institutions for AI adoption. AI's deployment in dental care can **widen the gap between developed and developing countries**. Low-income countries often lack the financial resources and information and communication technology infrastructure needed to implement AI, which can leave them further behind in healthcare advancements. Dentists in larger clinics with more modern equipment in high-income countries are more likely to implement and benefit from AI systems. This creates a disparity in incentives and benefits from AI between practitioners in different economic contexts. There is a risk that patients in low-income countries will experience contrasting healthcare services compared to those in high-income countries especially when it comes to AI-enhanced dental care. This inequality can represent a threat to the right to equal provision of healthcare as the benefits of AI might not be equally accessible to all [15].

6. Challenges for Dental AI Startups in Keeping Up with AI Advancements

Dental AI startups often face multiple challenges that can lead to their failure. One primary issue is the limited availability of resources, accessibility, and comprehensiveness of data, which hampers the development of robust AI models. The performance of deep learning models is highly dependent on large and diverse datasets to capture intrinsic statistical patterns and structures in dental imagery. However, many dental AI projects suffer from small datasets that do not fully exploit the positive correlation between dataset size and AI performance. Also, there is a lack of standardized research methodologies which further limits the impact and reliability of these AI solutions.

As AI technology advances so do the ethical considerations and guidelines that govern its use. Keeping up with these guidelines is challenging for dental AI startups. Many young companies struggle to comply with these evolving standards, which can hinder their growth and acceptance in the markets simply because of operating with limited resources, including funding and manpower.

V. Conclusion

Through a systematic literature review, this study identified several key findings and ethical dilemmas for the application of AI in dentistry. AI models, particularly those utilizing deep learning (DL) and convolutional neural networks (CNNs), have demonstrated superior performance compared to traditional diagnostic methods. These technologies excel in detecting dental conditions such as caries by swiftly analyzing radiographs and suggesting treatment plans. AI-enhanced diagnostic tools and applications have significantly decreased treatment durations for patients, leading to faster and more efficient dental procedures, enhancing patient convenience and satisfaction. Systems like Diagnocat have streamlined the diagnostic process and improved precision in dental charting, contributing to more personalized and predictive dental care. AI technologies have proven effective in reducing mean discrepancies in treatment planning and shortening treatment durations by 20%, as evidenced by randomized controlled trials.

In pediatric dental treatment planning, tools like Diagnocat have successfully segmented supernumerary teeth within minutes, showcasing rapid and accurate diagnostic capabilities. However, studies comparing AI-driven programs like CephX, WebCeph, and AudaxCeph highlight the need for standardization across AI platforms due to significant discrepancies in cephalometric analysis results.

AI also enhances patient engagement and compliance through tools like AI-based speech analysis and automated treatment planners, which reduce errors and examination times. Telehealth applications, such as Dental Monitoring (DM), have streamlined orthodontic treatment processes by reducing the number of in-person appointments and improving patient compliance through daily automated assessments. AI-powered intelligent toothbrushes and sensors provide real-time feedback on brushing techniques, although challenges remain in ensuring long-term compliance and image quality.

Studies show that dental students generally have positive perceptions of AI's potential, supporting its use in diagnostics and treatment planning and advocating for its inclusion in dental training. Large Language Models (LLMs) like ChatGPT have shown potential in enhancing dental education and patient care, though their integration must be approached with caution to avoid overreliance and ensure the development of critical analytical and decision-making skills in dental professionals. AI acts as a "Digital Assistant" in dentistry, enhancing tasks such as image analysis, appointment scheduling, and treatment planning. Dental assistants and nurses play a crucial role in reviewing AI-generated reports for accuracy, identifying relevant information, reporting errors, and communicating findings to patients. This collaboration ensures the effective integration of AI into clinical practice, highlighting the importance of human oversight. Replicating human judgment in orthodontic AI systems remains a complex challenge due to the nuanced nature of clinical decision-making. AI tools need to be customizable and adaptable to individual dentist preferences, allowing for easy modifications of AI suggestions.

The potential for medical malpractice significantly influences the adoption of AI in dental practice, necessitating critical evaluation of AI recommendations by dentists. Ethical issues arise when AI models prioritize aesthetic outcomes over comprehensive dental treatment. Ensuring ethical AI development involves designing systems that balance aesthetic and functional dental needs without deceptive influences. To fully understand the impact of AI on patient outcomes, comprehensive longitudinal studies are required. While most studies focus

on immediate accuracy and efficiency, long-term studies are essential to track patient health, identify long-term effects, and assess the real-world efficacy of AI-driven interventions in improving dental health. In conclusion, the integration of AI in dentistry holds great promise for enhancing diagnostic accuracy, treatment planning, patient engagement, and overall dental care. However, careful consideration of ethical implications, customization to individual needs, and rigorous long-term studies are necessary to maximize the benefits of AI while mitigating potential risks.

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